

Loops and comments

Computational Methods I

Course WBPH064-05

Lecture 2



university of
 groningen

faculty of science
and engineering

zernike institute for
advanced materials

jagoda.slawinska@rug.nl

Good news - we have five more TAs in our team

Ph 9, Ph 10, Ph 11, APH 1	Akram Abbasniabazeh Stephanie Gonzalez Migoni Alex Mednicov Josh Edmundson	Tuesday 13.00 - 15.00 Wednesday 11.00 - 13.00
APH 2, APH 3, APH 4, MA 1, MA 2	Markuss Kenins Sergio Alvarruiz Campos Harish Kumar Maurya	Wednesday 13.00 - 15.00 Thursday 13.00 - 15.00
Ph 1, Ph 2, Ph 3, Ph 4	Federico Loi Ahmad Dibajeh Harshan Madeshwaran	Wednesday 15.00 - 17.00 Friday 11.00 - 13.00
Ph 5, Ph 6, Ph 7, Ph 8	Gijsbert ten Hoven Daniel Sade Seppe Verheyen	Wednesday 17.00 - 19.00 Thursday 11.00 - 13.00



**university of
 groningen**

**faculty of science
and engineering**

**zernike institute for
advanced materials**

jagoda.slawinska@rug.nl

Short feedback

- How was your experience at the tutorials?
 - How was your experience working with a notebook?
 - Do you already see possible applications?
-
- If you haven't enrolled to a group, please come to the tutorial and ask TAs to enroll you
 - Please look at the information on Brightspace before sending an email
 - We cannot always respond to your email immediately
 - Please ask questions on the content of exercises at the tutorials



**university of
 groningen**

**faculty of science
 and engineering**

**zernike institute for
 advanced materials**

jagoda.slawinska@rug.nl

Today's outline

1. How to improve the legibility of your code by commenting
2. The basic rules about code style
3. How to use loops to reduce redundancy in your code;
4. Difference between while-loops and for-loops;
5. List comprehensions
6. Control statements



**university of
 groningen**

**faculty of science
 and engineering**

**zernike institute for
 advanced materials**

jagoda.slawinska@rug.nl

Why are the comments needed?

Comments start with the hash # character and you will see it everywhere!

```
1  #This comment will be ignored by the interpreter
2  print ("Comment Example")
3  |
```

- To make the structure of the code clearer
- So that I can understand the code after some time
- So that others can understand the code
- ...



university of
 groningen

faculty of science
and engineering

zernike institute for
advanced materials

jagoda.slawinska@rug.nl

Good and bad examples

```
1 for i in range(1, 21): # For every number from 1 up to and including 20:
2     if i % 3 == 0: # If the number is divisible by 3:
3         if i % 5 == 0: # And the number is divisible by 5:
4             print("FizzBuzz") # Let the program print "FizzBuzz".
5         else: # If the number is only divisible by 3:
6             print("Fizz") # Let the program print "Fizz".
7     elif i % 5 == 0: # If the number is only divisible by 5:
8         print("Buzz") # Let the program print "Buzz".
9     else: # If the number is not divisible by 3 or 5:
10        print(i) # Let the program print the number.
```



Good and bad examples

```
1 # Print all numbers from 1 up to and including 20.
2 # Replace multiples of 3 by "Fizz".
3 # Replace multiples of 5 by "Buzz".
4 # Replace multiples of both 3 and 5 by "Fizzbuzz".
5 for i in range(1, 21):
6     if i % 3 == 0:
7         if i % 5 == 0:
8             print("FizzBuzz")
9         else:
10            print("Fizz")
11    elif i % 5 == 0:
12        print("Buzz")
13    else:
14        print(i)
```



Bad comments due to bad variable naming


NO

```
T_F = [0, -10, 100, 20] # Temperatures in degrees Fahrenheit.  
T_C = [] # List that will contain temperatures in degrees Celsius.  
a = 1.8 # Conversion factor.  
b = -32 # Additional constant.  
for t in T_F:  
    t2 = (t + b)/a # Convert temperature from Fahrenheit to Celsius.  
    T_C.append(t2)  
print(T_C)
```


NO


YES

```
temps_fahrenheit = [0, -10, 100, 20]  
temps_celsius = []  
conversion_factor = 1.8  
conversion_offset = -32  
for fahrenheit in temps_fahrenheit:  
    celsius = (fahrenheit + conversion_offset)/conversion_factor  
    temps_celsius.append(celsius)  
print(temps_celsius)
```


YES



What makes the code ... nice?

- Clear and logical structure, good strategy, consistency
- Efficient methods and appropriate use of built-in functions
- Visual layout following PEP 8 style guidelines (e.g. indentation, blank lines, etc.)
- Not too few and not too many comments
- Reasonable naming conventions
- Appropriate documentation

PEP8 is the standard style guide for Python code.

<https://peps.python.org/pep-0008/>



**university of
 groningen**

**faculty of science
 and engineering**

**zernike institute for
 advanced materials**

jagoda.slawinska@rug.nl

Documenting functions via docstrings

```
def add(a, b):  
    """Add the numbers a and b."""  
    return a + b
```

```
help(add)
```

Help on function add in module __main__:

```
add(a, b)  
    Add the numbers a and b.
```

Always use docstring when you define a function
Starting week 3 mandatory to use in all exercises

<https://peps.python.org/pep-0257/>



university of
 groningen

faculty of science
 and engineering

zernike institute for
 advanced materials

jagoda.slawinska@rug.nl

Why do we need loops? For-loops and while-loops

We need a loop when we want to repeat something.
They are very important !!!

```
x_1 = 1*1  
x_2 = 2*2  
x_3 = 3*3  
x_4 = 4*4  
x_5 = 5*5
```

```
i=0  
while True:  
    i = i+1  
    print('This is line', i)
```

X
NO

```
This is line 16046  
This is line 16047  
This is line 16048  
This is line 16049  
This is line 16050  
This is line 16051  
This is line 16052  
This is line 16053  
This is line 16054
```

```
for i in range(1,11):  
    print('This is line', i)
```

✓
YES

```
This is line 1  
This is line 2  
This is line 3  
This is line 4  
This is line 5  
This is line 6  
This is line 7  
This is line 8  
This is line 9  
This is line 10
```



university of
 groningen

faculty of science
and engineering

zernike institute for
advanced materials

jagoda.slawinska@rug.nl

For-loops vs while-loops

While-loop:

- Condition checking similar to the **if**
- Repeats as long as the condition is satisfied
- Unsure how many steps are needed
- Counter (as complicated as you want)

For-loop:

- Does something n times
- I know how many steps I need
- To check a condition you need the **if**
- Convenient to loop over the list



university of
 groningen

faculty of science
 and engineering

zernike institute for
 advanced materials

jagoda.slawinska@rug.nl

List comprehensions

- Fast way to create a list with a for-loop
- Condensed way of coding
- Can contain conditionals

Structure:

- square bracket outside
- a for-loop inside the brackets

```
x = [i for i in range(1, 11)]  
print(x)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
x = [i for i in range(1, 11)]  
y = [i**2 for i in range(1, 11)]  
print(x)  
print(y)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```



Control statements

The execution of a loop can be controlled by:

- **continue** (returns to the top of the loop)
- **break** (ends the loop)
- **pass** (prevents indentation errors)

which are called control statements.

You can use them but please make sure that you know what you are doing

```
word = 'A_word' # Some string.  
if word.startswith('p'):  
    print("Might be peanuts.")  
if word.endswith('s'):  
    pass # Must be still coded  
if word == "peanuts":  
    pass
```



university of
 groningen

faculty of science
and engineering

zernike institute for
advanced materials

jagoda.slawinska@rug.nl

Control statements

Example of the use of continue statement
Can we write this code without 'continue'?

```
numbers=list(range(10))
print(numbers)
total=0
for number in numbers:
    if number % 2 == 0:
        continue
    total += number
print('The sum of odd numbers is', total, '.')
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
The sum of odd numbers is 25 .
```

Example of the use of break statement
Can we write this code without 'break'?

```
name = 'Long-name'
position = 0
for character in name:
    position = position+1
    if character == 'n':
        break # Pass control
if position < len(name)+1:
    print("1st n occurs at place ", position)
else:
    print('No letter n in the name')
```



university of
 groningen

faculty of science
and engineering

zernike institute for
advanced materials

jagoda.slawinska@rug.nl

TELL US ABOUT YOUR COURSE PROGRAMME!

Check your student mail and
take the National Student Survey!



LOST YOUR MAIL?
SCAN THE CODE!

